

$$b) \quad y(n) = x(n-4) + x(n-3) + x(n-2) + x(n-1) + x(n)$$

$$Y(z) = X(z) \cdot z^{-4} + X(z) z^{-3} + X(z) z^{-2} + X(z) z^{-1} + X(z) z^0$$

$$T(z) = z^{-4} + z^{-3} + z^{-2} + z^{-1} + 1 = \frac{1 + z + z^2 + z^3 + z^4}{z^4}$$

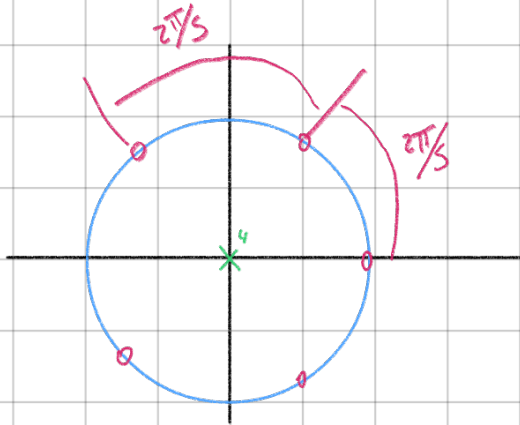
$\frac{z^5 - 1}{z - 1} \rightarrow \text{los ceros en } \frac{2\pi}{5}n; n \leq 5$

$$T(e^{j\omega}) = e^{-4j\omega} + e^{-3j\omega} + e^{-2j\omega} + e^{-j\omega} + 1 = e^{-2j\omega} \left(e^{-2j\omega} + e^{-j\omega} + 1 + e^{j\omega} + e^{2j\omega} \right)$$

$2\cos(\omega)$
 $2\cos(2\omega)$

$$T(e^{j\omega}) = e^{-2j\omega} (1 + 2\cos(\omega) + 2\cos(2\omega))$$

$$|T| = 1 + 2\cos(\omega) + 2\cos(2\omega) \quad \angle T = -2\omega$$



$$\begin{aligned} \omega = 0 & \quad |T| = 5 \\ |T| = 0 & \quad \begin{cases} \omega = \frac{1,2566}{\pi} = 0,4 \\ \omega = \frac{2,5132}{\pi} = 0,8 \end{cases} \end{aligned}$$